

Quadratic Reciprocity Law

§1 Quadratic Residues

1.1 Definition

Let $\gcd(a, m) = 1$. Then a is a *quadratic residue mod m* if the congruence $x^2 \equiv a \pmod{m}$ has a solution. Otherwise a is a *quadratic nonresidue mod m* .

1.2 Two Basic Problems

Two basic problems dominate the theory of quadratic residues:

- (a) Given a prime p , determine which n are quadratic residues modulo p and which are quadratic nonresidues modulo p .
- (b) Given n , determine those primes p for which n is a quadratic residue modulo p and those for which n is a quadratic nonresidue modulo p .

1.1 Proposition

Let p be an odd prime. Then every reduced residue system modulo p contains exactly $\frac{p-1}{2}$ quadratic residues and exactly $\frac{p-1}{2}$ quadratic nonresidues modulo p . The quadratic residues belong to the residue classes containing the integers $1^2, 2^2, \dots, \left(\frac{p-1}{2}\right)^2$.

PROOF

First observe that the integers $1^2, 2^2, \dots, \left(\frac{p-1}{2}\right)^2$ are distinct mod p . In fact, if $x^2 \equiv y^2 \pmod{p}$ with $1 \leq x, y \leq \frac{p-1}{2}$, then $(x-y)(x+y) \equiv 0 \pmod{p}$. But $1 < x+y < p$ so $x-y \equiv 0 \pmod{p}$. Hence, $x = y$. Since $(p-k)^2 \equiv k^2 \pmod{p}$, every quadratic residue is congruent mod p to exactly one of the integers $1^2, 2^2, \dots, \left(\frac{p-1}{2}\right)^2$.

1.2 Example

To find the quadratic residues modulo 11, square the integers 1, 2, ..., 10 and reduce modulo 11. Hence

$$1^2 \equiv 1, 2^2 \equiv 4, 3^2 \equiv 9, 4^2 \equiv 5, 5^2 \equiv 3 \pmod{11}.$$

According to (1.1), it is enough to square only the first half of the integers. In fact,

$$6^2 \equiv (-5)^2 \equiv 3, 7^2 \equiv (-4)^2 \equiv 5, 8^2 \equiv (-3)^2 \equiv 9, 9^2 \equiv (-2)^2 \equiv 4, 10^2 \equiv 1 \pmod{11}.$$

Therefore, the quadratic residues modulo 11 are 1, 3, 4, 5, 9 and the quadratic nonresidues are 2, 6, 7, 8, 10.

§2 Legendre's Symbol

2.1 Definition

Let p be an odd prime and $p \nmid a$.

Legendre's symbol $\left(\frac{a}{p}\right)$ is defined as follows:

$$\left(\frac{a}{p}\right) = \begin{cases} +1 & \text{if } a \text{ is a quadratic residue mod } p, \\ -1 & \text{if } a \text{ is a quadratic nonresidue mod } p. \end{cases}$$

If $p \mid a$, define $\left(\frac{a}{p}\right) = 0$.

2.2 Remarks

Here are some useful properties of the Legendre's symbol:

(a) The number of solutions to the $x^2 \equiv a \pmod{p}$, where p is an odd prime, is $1 + \left(\frac{a}{p}\right)$.

(b) (i) $\left(\frac{1}{p}\right) = 1$,

(ii) $\left(\frac{a^2}{p}\right) = 1$ for $p \nmid a$.

(c) If $a \equiv b \pmod{p}$, then

$$\left(\frac{a}{p}\right) = \left(\frac{b}{p}\right).$$

Therefore $\left(\frac{a}{p}\right)$ is a periodic function of a with period p .

(d) Using (1.1), it is clear that

$$\sum_{k=1}^{p-1} \left(\frac{k}{p}\right) = 0, \text{ where } p \text{ is an odd prime.}$$

2.3 Query

Fermat's Little Theorem tells us that $a^{p-1} \equiv 1 \pmod{p}$ if $p \nmid a$.

Since

$$a^{p-1} - 1 = \left(a^{\frac{p-1}{2}} - 1\right) \left(a^{\frac{p-1}{2}} + 1\right)$$

it follows that $a^{\frac{p-1}{2}} \equiv \pm 1 \pmod{p}$.

Is $a^{\frac{p-1}{2}}$ related to $\left(\frac{a}{p}\right)$? Yes, the next proposition answers this query.

2.4 Proposition (Euler's Criterion)

Let p be an odd prime. Then, for all a ,

$$\left(\frac{a}{p}\right) \equiv a^{\frac{p-1}{2}} \pmod{p}.$$

PROOF

If $p \mid a$, the result is trivial since both members are congruent to 0 mod p . Now suppose that $\left(\frac{a}{p}\right) = 1$. Then there is an x such that $x^2 \equiv a \pmod{p}$ and hence

$$a^{\frac{p-1}{2}} \equiv (x^2)^{\frac{p-1}{2}} = x^{p-1} \equiv 1 = \left(\frac{a}{p}\right) \pmod{p}.$$

Now suppose that $\left(\frac{a}{p}\right) = -1$ and consider the polynomial

$$f(x) = x^{\frac{p-1}{2}} - 1.$$

Since $f(x)$ has degree $\frac{p-1}{2}$, it follows that the congruence $f(x) \equiv 0 \pmod{p}$ has at most $\frac{p-1}{2}$ solutions (see Appendix). But the $\frac{p-1}{2}$ quadratic residues modulo p are solutions so the nonresidues are not. Hence $a^{\frac{p-1}{2}} \not\equiv 1 \pmod{p}$ if $\left(\frac{a}{p}\right) = -1$. But $a^{\frac{p-1}{2}} \equiv \pm 1 \pmod{p}$, so $a^{\frac{p-1}{2}} \equiv -1 = \left(\frac{a}{p}\right) \pmod{p}$.

2.5 Proposition

Let p be an odd prime. Then $\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right)\left(\frac{b}{p}\right)$.

PROOF

If $p \mid a$ or $p \mid b$, then $p \mid ab$ so $\left(\frac{ab}{p}\right) = 0$ and either $\left(\frac{a}{p}\right) = 0$ or $\left(\frac{b}{p}\right) = 0$. Thus $\left(\frac{ab}{p}\right) = \left(\frac{a}{p}\right)\left(\frac{b}{p}\right)$, if $p \mid a$ or $p \mid b$. If $p \nmid a$ and $p \nmid b$, then $p \nmid ab$. Thus

$$\left(\frac{ab}{p}\right) \equiv (ab)^{\frac{p-1}{2}} = a^{\frac{p-1}{2}} b^{\frac{p-1}{2}} \equiv \left(\frac{a}{p}\right)\left(\frac{b}{p}\right) \pmod{p}.$$

But each of $\left(\frac{ab}{p}\right)$, $\left(\frac{a}{p}\right)$, $\left(\frac{b}{p}\right)$ is either 1 or -1 so their difference $\left(\frac{ab}{p}\right) - \left(\frac{a}{p}\right)\left(\frac{b}{p}\right)$ is either 0, 2 or -2 . Since this difference is divisible by p , it must be 0.

2.6 Remark

- (a) (2.5) states that Legendre's symbol $\left(\frac{a}{p}\right)$ is a completely multiplicative function of a .
- (b) The product of two quadratic residues is a quadratic residue, the product of two quadratic nonresidues is a quadratic residue, and the product of a quadratic residue and a quadratic nonresidue is a quadratic nonresidue.

2.7 Proposition

For every odd prime p ,

$$\left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}} = \begin{cases} +1 & \text{if } p \equiv 1 \pmod{4}, \\ -1 & \text{if } p \equiv 3 \pmod{4}. \end{cases}$$

PROOF

By *Euler's Criterion*,

$$\left(\frac{-1}{p}\right) \equiv (-1)^{\frac{p-1}{2}} \pmod{p}.$$

Since each member of this congruence is 1 or -1 the two members are equal.

2.8 Proposition

For every odd prime p ,

$$\left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}} = \begin{cases} +1 & \text{if } p \equiv \pm 1 \pmod{8}, \\ -1 & \text{if } p \equiv \pm 3 \pmod{8}. \end{cases}$$

PROOF

Consider the following $\frac{p-1}{2}$ congruences mod p :

$$\begin{aligned} p-1 &\equiv 1(-1)^1 \pmod{p} \\ 2 &\equiv 2(-1)^2 \pmod{p} \\ p-3 &\equiv 3(-1)^3 \pmod{p} \\ 4 &\equiv 4(-1)^4 \pmod{p} \\ &\vdots \\ r &\equiv \frac{p-1}{2}(-1)^{\frac{p-1}{2}} \pmod{p} \end{aligned}$$

where r is either $p - \frac{p-1}{2}$ or $\frac{p-1}{2}$. Multiply these congruences together and note that each integer on the left is even. Thus,

$$2 \cdot 4 \cdot 6 \cdot \dots \cdot (p-1) \equiv \left(\frac{p-1}{2}\right)! (-1)^{1+2+\dots+\frac{p-1}{2}} \pmod{p}.$$

Consequently,

$$2^{\frac{p-1}{2}} \left(\frac{p-1}{2}\right)! \equiv \left(\frac{p-1}{2}\right)! (-1)^{\frac{p^2-1}{8}} \pmod{p}.$$

Now $p \nmid \left(\frac{p-1}{2}\right)!$ and so

$$2^{\frac{p-1}{2}} \equiv (-1)^{\frac{p^2-1}{8}} \pmod{p}.$$

By *Euler's Criterion*,

$$2^{\frac{p-1}{2}} \equiv \left(\frac{2}{p}\right) \pmod{p}$$

and since each member of this congruence is 1 or -1 the two members are equal.

2.9 Proposition (Gauss' Lemma)

For any odd prime p , with $p \nmid a$, consider $a, 2a, 3a, \dots, \left(\frac{p-1}{2}\right)a$ and their least positive residues mod p .

If ν is the number of residues which exceed $\frac{p}{2}$, then $\left(\frac{a}{p}\right) = (-1)^\nu$.

PROOF

The integers $a, 2a, 3a, \dots, \left(\frac{p-1}{2}\right)a$ are incongruent mod p . Consider their least positive residues and distribute them into two disjoint sets A and B , according as the residues are less than $\frac{p}{2}$ or greater than $\frac{p}{2}$. Thus $A = \{a_1, a_2, \dots, a_\mu\}$, where each $a_i \equiv ma \pmod{p}$ for some $m \leq \frac{p-1}{2}$ and $0 < a_i < \frac{p}{2}$; and $B = \{b_1, b_2, \dots, b_\nu\}$, where each $b_i \equiv na \pmod{p}$ for some $n \leq \frac{p-1}{2}$ and $\frac{p}{2} < b_i < p$. Notice that $\mu + \nu = \frac{p-1}{2}$ since A and B are disjoint. Form a new set C of ν elements by subtracting each b_i from p . Hence $C = \{c_1, c_2, \dots, c_\nu\}$, where $c_i = p - b_i$. Now $0 < c_i < \frac{p}{2}$ so the elements of C lie in the same interval as the elements of A . Suppose $a_i = c_j$ for some i, j . Then $a_i = p - b_j$ and so $a_i + b_j \equiv 0 \pmod{p}$. Thus $ma + na = (m+n)a \equiv 0 \pmod{p}$ for some m, n with $1 \leq m, n < \frac{p}{2}$. But this is impossible since $p \nmid a$ and $0 < m+n < p$. So A and C are disjoint and the union of A and C contains $\mu + \nu = \frac{p-1}{2}$ integers ranging from 1 to $\frac{p-1}{2}$. Thus $a_1 a_2 \dots a_\mu \cdot c_1 c_2 \dots c_\nu = \left(\frac{p-1}{2}\right)!$. Since $c_i = p - b_i$, then

$$\begin{aligned} \left(\frac{p-1}{2}\right)! &= a_1 a_2 \dots a_\mu (p - b_1)(p - b_2) \dots (p - b_\nu) \\ &\equiv (-1)^\nu a_1 a_2 \dots a_\mu b_1 b_2 \dots b_\nu \pmod{p} \\ &\equiv (-1)^\nu a(2a) \dots \left(\frac{p-1}{2}a\right) \pmod{p} \\ &\equiv (-1)^\nu a^{\frac{p-1}{2}} \left(\frac{p-1}{2}\right)! \pmod{p} \end{aligned}$$

Since $p \nmid \left(\frac{p-1}{2}\right)!$, cancel $\left(\frac{p-1}{2}\right)!$ from both sides. Thus $a^{\frac{p-1}{2}} \equiv (-1)^\nu \pmod{p}$. By *Euler's Criterion*, $\left(\frac{a}{p}\right) \equiv (-1)^\nu \pmod{p}$. Therefore $\left(\frac{a}{p}\right) = (-1)^\nu$.

2.10 Remarks

Although *Euler's Criterion* gives a straightforward method for computing $\left(\frac{a}{p}\right)$, the calculation may become prohibitive for large a since it requires raising a to the power $\frac{p-1}{2}$. *Gauss' Lemma* provides another criterion which involves a simpler calculation: determining the parity of ν .

2.11 Proposition

For any odd prime p , with $p \nmid a$, consider $a, 2a, 3a, \dots, \left(\frac{p-1}{2}\right)a$ and v be the number of their least positive residues mod p which exceed $\frac{p}{2}$. Then

$$v \equiv \sum_{k=1}^{\frac{p-1}{2}} \left[\frac{ka}{p} \right] + (a-1) \frac{p^2-1}{8} \pmod{2}.$$

PROOF

For each of $k = 1, 2, 3, \dots, \frac{p-1}{2}$, observe that

$$\frac{ka}{p} = \left[\frac{ka}{p} \right] + \left\{ \frac{ka}{p} \right\}, \text{ where } 0 < \left\{ \frac{ka}{p} \right\} < 1.$$

Now, let $r_k = ka - p \left[\frac{ka}{p} \right]$, which is the least positive residue of $ka \pmod{p}$. Refer to the sets used in the proof of *Gauss' Lemma* and get $\{r_1, r_2, \dots, r_{\frac{p-1}{2}}\} = \{a_1, a_2, \dots, a_\mu, b_1, b_2, \dots, b_v\}$. Moreover, recall also that $\{1, 2, 3, \dots, \frac{p-1}{2}\} = \{a_1, a_2, \dots, a_\mu, c_1, c_2, \dots, c_v\}$, where $c_i = p - b_i$.

Compute the sums of the elements in these sets to obtain two equations:

$$\sum_{k=1}^{\frac{p-1}{2}} r_k = \sum_{i=1}^{\mu} a_i + \sum_{i=1}^v b_i$$

and

$$\sum_{k=1}^{\frac{p-1}{2}} k = \sum_{i=1}^{\mu} a_i + \sum_{i=1}^v c_i = \sum_{i=1}^{\mu} a_i + vp - \sum_{i=1}^v b_i.$$

Hence, from the first equation,

$$\sum_{i=1}^{\mu} a_i + \sum_{i=1}^v b_i = a \sum_{k=1}^{\frac{p-1}{2}} k - p \sum_{k=1}^{\frac{p-1}{2}} \left[\frac{ka}{p} \right].$$

Add this to the second equation and get,

$$vp + 2 \sum_{i=1}^{\mu} a_i = (a+1) \sum_{k=1}^{\frac{p-1}{2}} k - p \sum_{k=1}^{\frac{p-1}{2}} \left[\frac{ka}{p} \right] = (a+1) \frac{p^2-1}{8} - p \sum_{k=1}^{\frac{p-1}{2}} \left[\frac{ka}{p} \right].$$

Reduce this mod 2, noting that $a+1 \equiv a-1 \pmod{2}$ and $p \equiv 1 \pmod{2}$, and obtain

$$v \equiv \sum_{k=1}^{\frac{p-1}{2}} \left[\frac{ka}{p} \right] + (a-1) \frac{p^2-1}{8} \pmod{2}.$$

2.12 Remark

In particular, if a is odd, then $v \equiv \sum_{k=1}^{\frac{p-1}{2}} \left[\frac{ka}{p} \right] \pmod{2}$.

§3 Quadratic Reciprocity Law

3.1 Theorem (Quadratic Reciprocity Law)

If p and q are distinct odd primes, then $\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}}$.

PROOF

By Gauss' Lemma and (2.11),

$$\left(\frac{q}{p}\right) = (-1)^v, \text{ where } v \equiv \sum_{m=1}^{\frac{p-1}{2}} \left[\frac{mq}{p}\right] \pmod{2},$$

and

$$\left(\frac{p}{q}\right) = (-1)^\mu, \text{ where } \mu \equiv \sum_{n=1}^{\frac{q-1}{2}} \left[\frac{np}{q}\right] \pmod{2}.$$

Hence $\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\mu+v}$, and it suffices to show that

$$\sum_{m=1}^{\frac{p-1}{2}} \left[\frac{mq}{p}\right] + \sum_{n=1}^{\frac{q-1}{2}} \left[\frac{np}{q}\right] = \frac{p-1}{2} \cdot \frac{q-1}{2}.$$

Consider the function

$$f(m, n) = mq - np, \text{ for } |m| < \frac{p}{2} \text{ and } |n| < \frac{q}{2}.$$

If m and n are nonzero integers then $f(m, n)$ is a nonzero integer.

As m takes the values $1, 2, 3, \dots, \frac{p-1}{2}$ and n takes the values $1, 2, 3, \dots, \frac{q-1}{2}$, then $f(m, n)$ takes $\frac{p-1}{2} \cdot \frac{q-1}{2}$ values, no two of which are equal because

$$f(m, n) - f(m', n') = f(m - m', n - n') \neq 0.$$

Now, for each fixed $m = 1, 2, 3, \dots, \frac{p-1}{2}$,

$$f(m, n) > 0 \text{ if, and only if } 1 \leq n < \frac{mq}{p}, \text{ or equivalently, } 1 \leq n \leq \left[\frac{mq}{p}\right].$$

Thus the total number of positive values is $\sum_{m=1}^{\frac{p-1}{2}} \left[\frac{mq}{p}\right]$.

Similarly, for each fixed $n = 1, 2, 3, \dots, \frac{q-1}{2}$,

$$f(m, n) < 0 \text{ if, and only if } 1 \leq m < \frac{np}{q}, \text{ or equivalently, } 1 \leq m \leq \left[\frac{np}{q}\right].$$

Thus the total number of negative values is $\sum_{n=1}^{\frac{q-1}{2}} \left[\frac{np}{q}\right]$.

Since the total number of positive and negative values together is $\frac{p-1}{2} \cdot \frac{q-1}{2}$, the result follows.

3.2 Remark

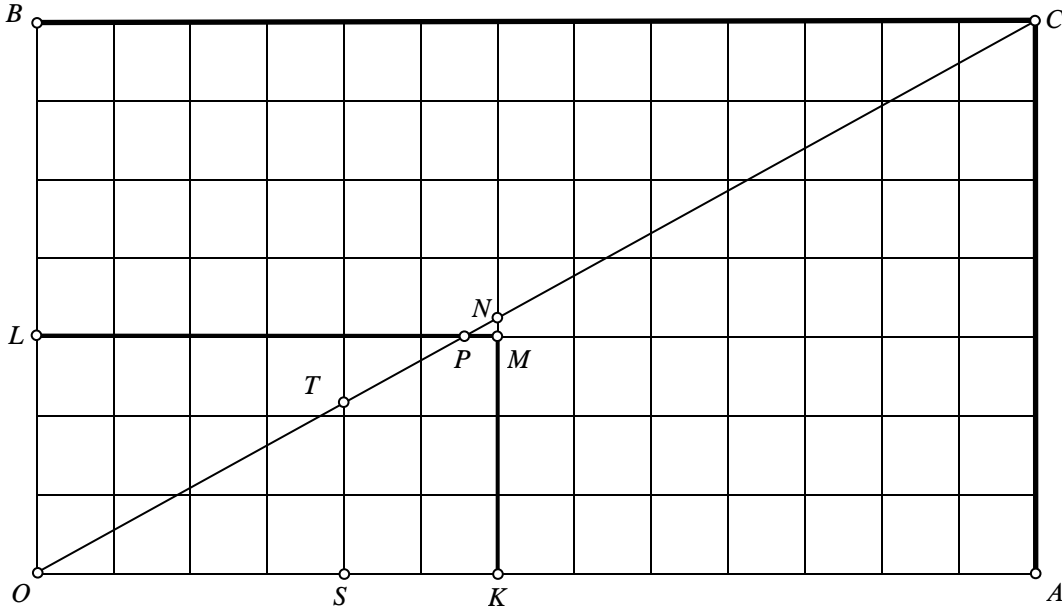
The first complete proof of the theorem (recently recast as a proof by descent) was due to Gauss. Now (up until 2003), at least 207 proofs of the quadratic reciprocity law have been published. See, for example:

<http://www.rzuser.uni-heidelberg.de/~hb3/rchrono.html>.

Gauss himself supplied no less than 8, including a version of the one just given.

3.3 Proof via Lattice Points

Here is how to recast the proof in (3.1) using lattice points.



In the figure, AC and BC are $x = p$, $y = q$, and KM and LM are $x = p'$, $y = q'$. If (as in the figure) $p > q$, then $\frac{q'}{p'} < \frac{q}{p}$ and M falls below the diagonal OC . Since $q' < \frac{qp'}{p} < q' + 1$, there is no integer between $KM = q'$ and $KN = \frac{qp'}{p}$.

Count up the lattice points in two different ways in rectangle $OKML$, counting the points on KM and LM but not those on the axes.

In the first place, this number is $p'q'$. But there are no lattice points on OC (since p and q are relatively prime), and none in $\triangle PMN$ except perhaps on PM . Hence the number of lattice points in $OKML$ is the sum of those in $\triangle OKN$ and $\triangle OLP$ (counting those on KN and LP but not those on the axes).

The number on ST , the line $x = s$, is $\left\lfloor \frac{sq}{p} \right\rfloor$. Hence the number of lattice points in $\triangle OKN$ is $\sum_{s=1}^{p'} \left\lfloor \frac{sq}{p} \right\rfloor$.

Similarly, the number of lattice points in $\triangle OLP$ is $\sum_{s=1}^{q'} \left\lfloor \frac{sp}{q} \right\rfloor$.

3.4 Lemma

If n is an odd positive integer, then $x^n - y^n = \prod_{k=0}^{n-1} (\omega^k x - \omega^{-k} y)$, where $\omega = e^{\frac{2\pi i}{n}}$.

PROOF

$1, \omega, \omega^2, \dots, \omega^{n-1}$ are all roots of the polynomial $z^n - 1$. Since there are n of them and they are all distinct, so $z^n - 1 = \prod_{k=0}^{n-1} (z - \omega^k)$. Let $z = \frac{x}{y}$. Multiply both sides by y^n and get $x^n - y^n = \prod_{k=0}^{n-1} (x - \omega^k y)$. Since n is odd, as k runs over a complete system of residues mod n , so does $-2k$. Thus

$$\begin{aligned} x^n - y^n &= \prod_{k=0}^{n-1} (x - \omega^{-2k} y) \\ &= \omega^{-(1+2+\dots+n-1)} \prod_{k=0}^{n-1} (\omega^k x - \omega^{-k} y) \\ &= \prod_{k=0}^{n-1} (\omega^k x - \omega^{-k} y). \end{aligned}$$

In the last step the fact that $1 + 2 + \dots + n - 1$ is divisible by n was used.

3.5 Lemma

Let n is an odd positive integer and let $f(z) = e^{2\pi iz} - e^{-2\pi iz}$. Then $\frac{f(nz)}{f(z)} = \prod_{k=1}^{\frac{n-1}{2}} f\left(z + \frac{k}{n}\right) f\left(z - \frac{k}{n}\right)$.

PROOF

Substitute $x = e^{2\pi iz}$ and $y = e^{-2\pi iz}$ in (3.4) and get $f(nz) = \prod_{k=0}^{n-1} f\left(z + \frac{k}{n}\right)$. Since f is periodic with period 1, then $f\left(z + \frac{k}{n}\right) = f\left(z + \frac{k}{n} - 1\right) = f\left(z - \frac{n-k}{n}\right)$. Thus

$$\begin{aligned} \frac{f(nz)}{f(z)} &= \prod_{k=1}^{\frac{n-1}{2}} f\left(z + \frac{k}{n}\right) \prod_{k=\frac{n+1}{2}}^{n-1} f\left(z + \frac{k}{n}\right) \\ &= \prod_{k=1}^{\frac{n-1}{2}} f\left(z + \frac{k}{n}\right) \prod_{k=\frac{n+1}{2}}^{n-1} f\left(z - \frac{n-k}{n}\right) \\ &= \prod_{k=1}^{\frac{n-1}{2}} f\left(z + \frac{k}{n}\right) f\left(z - \frac{k}{n}\right). \end{aligned}$$

Note that as k runs from $\frac{n+1}{2}$ to $n-1$, $n-k$ runs from $\frac{n-1}{2}$ to 1.

3.6 Lemma

Let p be an odd prime, $p \nmid a$ and let $f(z) = e^{2\pi iz} - e^{-2\pi iz}$. Then $\prod_{k=1}^{\frac{p-1}{2}} f\left(\frac{ka}{p}\right) = \left(\frac{a}{p}\right) \prod_{k=1}^{\frac{p-1}{2}} f\left(\frac{k}{p}\right)$.

PROOF

Note that $ka \equiv \pm r_k \pmod{p}$ where $1 \leq r_k \leq \frac{p-1}{2}$. Thus $\frac{ka}{p}$ and $\pm \frac{r_k}{p}$ differ by an integer. This implies that $f\left(\frac{ka}{p}\right) = f\left(\pm \frac{r_k}{p}\right) = \pm f\left(\frac{r_k}{p}\right)$. The result now follows by taking the product of both sides as k runs from 1 to $\frac{p-1}{2}$ and applying Gauss' Lemma.

3.7 Historical Note

Below is another ingenious proof of the *Quadratic Reciprocity Law* due to Eisenstein found in his paper *Applications de l'Algebre a l'Arithmetique transcendante*. Kummer, in a historical study of the laws of reciprocity, refers to this proof as one of the most beautiful of all proofs ("...einen der schönsten Beweise dieses von den ausgezeichnetsten Mathematikern viel bewiesenen Theorems..."). Replacing the trigonometric function by certain elliptic functions, Eisenstein was able, without much more difficulty, to prove the laws of cubic and biquadratic reciprocity as well.

3.8 Another Proof of the Quadratic Reciprocity Law

Let p and q be distinct odd primes and let $f(z) = e^{2\pi iz} - e^{-2\pi iz}$.

Then, by (3.6),

$$\prod_{m=1}^{\frac{p-1}{2}} f\left(\frac{mq}{p}\right) = \left(\frac{q}{p}\right) \prod_{m=1}^{\frac{p-1}{2}} f\left(\frac{m}{p}\right).$$

By (3.5),

$$\frac{f\left(\frac{mq}{p}\right)}{f\left(\frac{m}{p}\right)} = \prod_{n=1}^{\frac{q-1}{2}} f\left(\frac{m}{p} + \frac{n}{q}\right) f\left(\frac{m}{p} - \frac{n}{q}\right).$$

Put these two equations together and get

$$\left(\frac{q}{p}\right) = \prod_{m=1}^{\frac{p-1}{2}} \prod_{n=1}^{\frac{q-1}{2}} f\left(\frac{m}{p} + \frac{n}{q}\right) f\left(\frac{m}{p} - \frac{n}{q}\right).$$

Similarly,

$$\left(\frac{p}{q}\right) = \prod_{n=1}^{\frac{q-1}{2}} \prod_{m=1}^{\frac{p-1}{2}} f\left(\frac{n}{q} + \frac{m}{p}\right) f\left(\frac{n}{q} - \frac{m}{p}\right).$$

Since $f\left(\frac{n}{q} - \frac{m}{p}\right) = -f\left(\frac{m}{p} - \frac{n}{q}\right)$, then

$$\begin{aligned} \left(\frac{p}{q}\right) &= \prod_{n=1}^{\frac{q-1}{2}} \prod_{m=1}^{\frac{p-1}{2}} f\left(\frac{n}{q} + \frac{m}{p}\right) f\left(\frac{n}{q} - \frac{m}{p}\right) \\ &= (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}} \prod_{n=1}^{\frac{q-1}{2}} \prod_{m=1}^{\frac{p-1}{2}} f\left(\frac{m}{p} + \frac{n}{q}\right) f\left(\frac{m}{p} - \frac{n}{q}\right) \\ &= (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}} \left(\frac{q}{p}\right). \end{aligned}$$

3.9 Remarks

The *Quadratic Reciprocity Law* can be restated as follows:

- (a). If either p or q is of the form $4n + 1$, then p is a quadratic residue mod q if, and only if q is a quadratic residue mod p .
- (b). If both p and q are of the form $4n + 3$, then p is a quadratic residue mod q if, and only if q is a quadratic nonresidue mod p .

3.10 Theorem

Let q be an odd prime.

- (a). If $q \equiv 1 \pmod{4}$, then q is a quadratic residue mod p if, and only if $p \equiv r \pmod{q}$, where r is a quadratic residue mod q .
- (b). If $q \equiv 3 \pmod{4}$, then q is a quadratic residue mod p if, and only if $p \equiv \pm b^2 \pmod{4q}$, where b is an odd integer relatively prime to q .

PROOF

- (a). If $q \equiv 1 \pmod{4}$, then by the *Quadratic Reciprocity Law*,

$$\left(\frac{q}{p}\right) = \left(\frac{p}{q}\right).$$

Thus the result follows.

- (b). If $q \equiv 3 \pmod{4}$, then by the *Quadratic Reciprocity Law*,

$$\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2}} \left(\frac{p}{q}\right).$$

Now, assume that $p \equiv \pm b^2 \pmod{4q}$, where b is an odd integer relatively prime to q .

If the plus sign is chosen then $p \equiv b^2 \equiv 1 \pmod{4}$ and $p \equiv b^2 \pmod{q}$.

Thus $(-1)^{\frac{p-1}{2}} = 1$ and $\left(\frac{p}{q}\right) = 1$ so $\left(\frac{q}{p}\right) = 1$.

If the minus sign is chosen, then

$$\left(\frac{p}{q}\right) = \left(\frac{-b^2}{q}\right) = \left(\frac{-1}{q}\right) \left(\frac{b}{q}\right)^2 = \left(\frac{-1}{q}\right) = -1$$

since $q \equiv 3 \pmod{4}$. Moreover, since $p \equiv -b^2 \equiv -1 \equiv 3 \pmod{4}$ so $(-1)^{\frac{p-1}{2}} = -1$.

Consequently, $\left(\frac{q}{p}\right) = 1$.

Now assume that $\left(\frac{q}{p}\right) = 1$.

Then either

- (i). $(-1)^{\frac{p-1}{2}} = 1$ and $\left(\frac{p}{q}\right) = 1$, or

- (ii). $(-1)^{\frac{p-1}{2}} = -1$ and $\left(\frac{p}{q}\right) = -1$.

In the first case, $p \equiv 1 \pmod{4}$ and $p \equiv b^2 \pmod{q}$, where b can be assumed to be odd since if it is even then, $b' = b + q$ can be used instead. If b is odd then $b^2 \equiv 1 \pmod{4}$ and $p \equiv b^2 \pmod{4}$ and thus $p \equiv b^2 \pmod{4q}$. In the second case, $p \equiv 3 \pmod{4}$ and $p \equiv -b^2 \pmod{q}$. The last congruence follows from the fact that $q \equiv 3 \pmod{4}$ implies that every quadratic nonresidue of q is the negative of its quadratic residue. Again b maybe assumed to be odd. In that case, $-b^2 \equiv 3 \pmod{4}$ so $p \equiv -b^2 \pmod{4}$ and $p \equiv -b^2 \pmod{4q}$.

3.11 Theorem

Let p and q distinct odd primes and let a be a positive integer. Then

$$\left(\frac{p}{q}\right)\left(\frac{q}{p}\right) = (-1)^{\frac{p-1}{2} \cdot \frac{q-1}{2}} \text{ if, and only if, } \left(\frac{a}{p}\right) = \left(\frac{a}{q}\right) \text{ whenever } p \equiv \pm q \pmod{4a}, p \nmid a.$$

PROOF

In order to show that (a) implies (b), it is enough, by multiplicativity, to show that (b) holds when a is prime. For $a = 2$, the result follows from (2.8). If a is an odd prime then by (a),

$$\left(\frac{a}{p}\right) = (-1)^{\frac{p-1}{2} \cdot \frac{a-1}{2}} \left(\frac{p}{a}\right).$$

If $p \equiv q \pmod{4a}$, then $\left(\frac{p}{a}\right) = \left(\frac{q}{a}\right)$ so that

$$\begin{aligned} \left(\frac{a}{p}\right) &= (-1)^{\frac{p-1}{2} \cdot \frac{a-1}{2}} \left(\frac{q}{a}\right) \\ &= (-1)^{\frac{p-1}{2} \cdot \frac{a-1}{2}} (-1)^{\frac{q-1}{2} \cdot \frac{a-1}{2}} \left(\frac{a}{q}\right) \\ &= (-1)^{\frac{a-1}{2} \cdot \frac{p+q-2}{2}} \left(\frac{a}{q}\right). \end{aligned}$$

But $p \equiv q \pmod{4a}$ implies $p + q - 2 \equiv 0 \pmod{4}$ and the result follows.

If, on the other hand, $p \equiv -q \pmod{4a}$, a similar calculation shows

$$\left(\frac{a}{p}\right) = (-1)^{\frac{a-1}{2} \cdot \frac{p+q}{2}} \left(\frac{a}{q}\right).$$

Since $p + q \equiv 0 \pmod{4}$, the result holds in this case.

To show that (b) implies (a), suppose first of all that $p > q$ and $p \equiv q \pmod{4}$.

Then $p = q + 4a$ for $a \geq 1$.

Thus

$$\left(\frac{p}{q}\right) = \left(\frac{q + 4a}{q}\right) = \left(\frac{a}{q}\right) = \left(\frac{a}{p}\right) = \left(\frac{4a}{p}\right) = \left(\frac{p - q}{p}\right) = \left(\frac{-q}{p}\right) = (-1)^{\frac{p-1}{2}} \left(\frac{q}{p}\right).$$

If $p \equiv 1 \pmod{4}$, then $\left(\frac{p}{q}\right) = \left(\frac{q}{p}\right)$ which gives (a).

If $p \equiv 3 \pmod{4}$, then $q \equiv 3 \pmod{4}$ and we get $\left(\frac{p}{q}\right) = -\left(\frac{q}{p}\right)$ which is (a) in that case.

Finally, if $p \equiv -q \pmod{4}$, then $p + q = 4a$ and

$$\left(\frac{p}{q}\right) = \left(\frac{-q + 4a}{q}\right) = \left(\frac{a}{q}\right) = \left(\frac{a}{p}\right) = \left(\frac{4a}{p}\right) = \left(\frac{p + q}{p}\right) = \left(\frac{q}{p}\right).$$

Thus $\left(\frac{p}{q}\right) = \left(\frac{q}{p}\right)$ which is (a) since in this case at least one of p or q must be congruent to 1 mod 4.

3.12 Remarks

(3.11) gives an equivalent formulation of the *Quadratic Reciprocity Law*.

§4 Quadratic Gauss Sum

4.1 Example

Let $\zeta = e^{\frac{2\pi i}{8}}$, the primitive eighth root of unity. Thus $0 = \zeta^8 - 1 = (\zeta^4 - 1)(\zeta^4 + 1)$. Since $\zeta^4 \neq 1$, then $\zeta^4 = -1$. Multiply both sides by $\frac{1}{\zeta^2}$ and get $\zeta^2 + \frac{1}{\zeta^2} = 0$.

Let $\tau = \zeta + \frac{1}{\zeta}$ and let p be an odd prime. Then $\tau^2 = \left(\zeta + \frac{1}{\zeta}\right)^2 = \zeta^2 + 2 + \frac{1}{\zeta^2} = 2$ and so

$$\tau^{p-1} = (\tau^2)^{\frac{p-1}{2}} = 2^{\frac{p-1}{2}} \equiv \left(\frac{2}{p}\right) \pmod{p}.$$

It follows that

$$\tau^p \equiv \left(\frac{2}{p}\right) \tau \pmod{p}.$$

Now,

$$(a). \quad \tau^p = \left(\zeta + \frac{1}{\zeta}\right)^p \equiv \zeta^p + \frac{1}{\zeta^p} \pmod{p}.$$

$$(b). \quad \zeta^p + \frac{1}{\zeta^p} = \zeta + \frac{1}{\zeta} = \tau \text{ if } p \equiv \pm 1 \pmod{8}.$$

$$(c). \quad \zeta^p + \frac{1}{\zeta^p} = \zeta^3 + \frac{1}{\zeta^3} = -\frac{1}{\zeta} + (-\zeta) = -\tau \text{ if } p \equiv \pm 3 \pmod{8}.$$

So $(-1)^{\frac{p^2-1}{8}} \tau \equiv \left(\frac{2}{p}\right) \tau \pmod{p}$. Multiply both sides by τ and get $(-1)^{\frac{p^2-1}{8}} 2 \equiv \left(\frac{2}{p}\right) 2 \pmod{p}$.

Therefore, $\left(\frac{2}{p}\right) \equiv (-1)^{\frac{p^2-1}{8}} \pmod{p}$.

4.2 Notation

Let $\zeta = e^{\frac{2\pi i}{p}}$, where p is an odd prime.

4.3 Lemma

$$\sum_{r=0}^{p-1} \zeta^{nr} = \begin{cases} p & \text{if } n \equiv 0 \pmod{p}, \\ 0 & \text{if } n \not\equiv 0 \pmod{p}. \end{cases}$$

PROOF

If $n \equiv 0 \pmod{p}$, then $\zeta^n = 1$ and so $\sum_{r=0}^{p-1} \zeta^{nr} = \sum_{r=0}^{p-1} 1 = p$.

If $n \not\equiv 0 \pmod{p}$, then $\zeta^n \neq 1$ and so $\sum_{r=0}^{p-1} \zeta^{nr} = \frac{\zeta^{np} - 1}{\zeta^n - 1} = 0$.

4.4 Lemma

$$\frac{1}{p} \sum_{r=0}^{p-1} \zeta^{(m-n)r} = \begin{cases} 1 & \text{if } m \equiv n \pmod{p}, \\ 0 & \text{if } m \not\equiv n \pmod{p}. \end{cases}$$

PROOF

If $m - n \equiv 0 \pmod{p}$, then $\zeta^{m-n} = 1$ and so $\frac{1}{p} \sum_{r=0}^{p-1} \zeta^{(m-n)r} = \frac{1}{p} \sum_{r=0}^{p-1} 1 = 1$.

If $m - n \not\equiv 0 \pmod{p}$, then $\zeta^{m-n} \neq 1$ and so $\frac{1}{p} \sum_{r=0}^{p-1} \zeta^{(m-n)r} = \frac{1}{p} \cdot \frac{\zeta^{(m-n)p} - 1}{\zeta^{m-n} - 1} = 0$.

4.5 Remark

Define

$$\delta(m, n) = \begin{cases} 1 & \text{if } m \equiv n \pmod{p}, \\ 0 & \text{if } m \not\equiv n \pmod{p}. \end{cases}$$

Then

$$\frac{1}{p} \sum_{r=0}^{p-1} \zeta^{(m-n)r} = \delta(m, n).$$

4.6 Definition

A *quadratic Gauss sum* g_a is defined as $g_a = \sum_{r=0}^{p-1} \left(\frac{r}{p}\right) \zeta^{ar}$.

4.7 Remarks

Since the sequence $1^2, 2^2, \dots, (p-1)^2$ contains each quadratic residue of p twice and contains no quadratic nonresidue

$$\sum_{r=0}^{p-1} \zeta^{r^2} = \sum_{r=0}^{p-1} \left(1 + \left(\frac{r}{p}\right)\right) \zeta^r = \sum_{r=0}^{p-1} \zeta^r + \sum_{r=0}^{p-1} \left(\frac{r}{p}\right) \zeta^r = \sum_{r=0}^{p-1} \left(\frac{r}{p}\right) \zeta^r.$$

Therefore, alternatively, $g_1 = \sum_{r=0}^{p-1} \zeta^{r^2}$.

4.8 Proposition

$$g_a = \left(\frac{a}{p}\right) g_1.$$

PROOF

Suppose $a \equiv 0 \pmod{p}$. Then $\zeta^{ar} = 1$ for all integers r and so,

$$g_a = \sum_{r=0}^{p-1} \left(\frac{r}{p}\right) \zeta^{ar} = \sum_{r=0}^{p-1} \left(\frac{r}{p}\right) = 0.$$

Suppose $a \not\equiv 0 \pmod{p}$. Then

$$\left(\frac{a}{p}\right) g_a = \sum_{r=0}^{p-1} \left(\frac{ar}{p}\right) \zeta^{ar} = \sum_{s=0}^{p-1} \left(\frac{s}{p}\right) \zeta^s = g_1,$$

because $s = ar$ runs over a complete residue system mod p when r does.

Multiply both sides by $\left(\frac{a}{p}\right)$ and get $g_a = \left(\frac{a}{p}\right) g_1$.

4.9 Proposition

$$g_1^2 = \left(\frac{-1}{p}\right) p = (-1)^{\frac{p-1}{2}} p.$$

PROOF

The idea of the proof is to evaluate $\sum_{a=0}^{p-1} g_a g_{-a}$ in two ways.

Suppose $a \not\equiv 0 \pmod{p}$. Then

$$g_a g_{-a} = \left(\frac{a}{p}\right) \left(\frac{-a}{p}\right) g_1^2 = \left(\frac{-1}{p}\right) g_1^2.$$

And so

$$\sum_{a=0}^{p-1} g_a g_{-a} = (p-1) \left(\frac{-1}{p} \right) g_1^2.$$

Now, notice that

$$g_a g_{-a} = \left(\sum_{r=0}^{p-1} \left(\frac{r}{p} \right) \zeta^{ar} \right) \left(\sum_{s=0}^{p-1} \left(\frac{s}{p} \right) \zeta^{-as} \right) = \sum_{r=0}^{p-1} \sum_{s=0}^{p-1} \left(\frac{r}{p} \right) \left(\frac{s}{p} \right) \zeta^{a(r-s)}.$$

Sum both sides over a and get

$$\sum_{a=0}^{p-1} g_a g_{-a} = \sum_{r=0}^{p-1} \sum_{s=0}^{p-1} \left(\frac{r}{p} \right) \left(\frac{s}{p} \right) \left(\sum_{a=0}^{p-1} \zeta^{a(r-s)} \right) = \sum_{r=0}^{p-1} \sum_{s=0}^{p-1} \left(\frac{r}{p} \right) \left(\frac{s}{p} \right) p \delta(r, s) = (p-1)p.$$

Put these two results together and get

$$(p-1) \left(\frac{-1}{p} \right) g_1^2 = (p-1)p.$$

Therefore

$$g_1^2 = \left(\frac{-1}{p} \right) p = (-1)^{\frac{p-1}{2}} p.$$

4.10 Remark

Another proof of (4.9) is to evaluate g_1^2 directly.

PROOF

Notice that

$$\begin{aligned} g_1^2 &= \left(\sum_{r=0}^{p-1} \left(\frac{r}{p} \right) \zeta^r \right) \left(\sum_{s=0}^{p-1} \left(\frac{s}{p} \right) \zeta^s \right) \\ &= \left(\sum_{r=1}^{p-1} \left(\frac{r}{p} \right) \zeta^r \right) \left(\sum_{s=1}^{p-1} \left(\frac{s}{p} \right) \zeta^s \right) \\ &= \sum_{r=1}^{p-1} \sum_{s=1}^{p-1} \left(\frac{r}{p} \right) \left(\frac{s}{p} \right) \zeta^{(r+s)}. \end{aligned}$$

For each pair r, s , there is a unique $t \bmod p$ such that $s \equiv tr \pmod{p}$, and

$$\left(\frac{r}{p} \right) \left(\frac{s}{p} \right) = \left(\frac{r}{p} \right) \left(\frac{tr}{p} \right) = \left(\frac{r}{p} \right) \left(\frac{t}{p} \right) \left(\frac{r}{p} \right) = \left(\frac{r}{p} \right)^2 \left(\frac{t}{p} \right) = \left(\frac{t}{p} \right).$$

Thus,

$$g_1^2 = \sum_{r=1}^{p-1} \sum_{s=1}^{p-1} \left(\frac{r}{p} \right) \left(\frac{s}{p} \right) \zeta^{(r+s)} = \sum_{t=1}^{p-1} \sum_{r=1}^{p-1} \left(\frac{t}{p} \right) \zeta^{r(1+t)} = \sum_{t=1}^{p-1} \left(\frac{t}{p} \right) \sum_{r=1}^{p-1} \zeta^{r(1+t)}.$$

Now,

$$\sum_{r=1}^{p-1} \zeta^{r(1+t)} = \begin{cases} -1 & \text{if } 1+t \not\equiv 0 \pmod{p}, \\ p-1 & \text{if } 1+t \equiv 0 \pmod{p}. \end{cases}$$

Therefore,

$$\begin{aligned}
g_1^2 &= \sum_{t=1}^{p-1} \left(\frac{t}{p} \right) \sum_{r=1}^{p-1} \zeta^{r(1+t)} \\
&= - \sum_{t=1}^{p-2} \left(\frac{t}{p} \right) + (p-1) \left(\frac{p-1}{p} \right) \\
&= - \sum_{t=1}^{p-1} \left(\frac{t}{p} \right) + p \left(\frac{p-1}{p} \right) \\
&= p \left(\frac{-1}{p} \right).
\end{aligned}$$

4.11 Yet Another Proof of the Quadratic Reciprocity Law

Let p and q be two distinct odd primes and set $p^* = (-1)^{\frac{p-1}{2}} p$.

Then

$$g_1^{q-1} = (g_1^2)^{\frac{q-1}{2}} = (p^*)^{\frac{q-1}{2}} \equiv \left(\frac{p^*}{q} \right) \pmod{q}.$$

Thus

$$g_1^q \equiv \left(\frac{p^*}{q} \right) g_1 \pmod{q}.$$

Notice that

$$g_1^q = \left(\sum_{r=0}^{p-1} \left(\frac{r}{p} \right) \zeta^r \right)^q \equiv \sum_{r=0}^{p-1} \left(\frac{r}{p} \right)^q \zeta^{rq} = \sum_{r=0}^{p-1} \left(\frac{r}{p} \right) \zeta^{rq} = g_q \pmod{q}.$$

By (4.8), it follows that

$$g_1^q \equiv g_q = \left(\frac{q}{p} \right) g_1 \pmod{q}.$$

And so

$$\left(\frac{q}{p} \right) g_1 \equiv \left(\frac{p^*}{q} \right) g_1 \pmod{q}.$$

Multiply both sides by g_1 and get

$$\left(\frac{q}{p} \right) p^* \equiv \left(\frac{p^*}{q} \right) p^* \pmod{q}$$

which implies that

$$\left(\frac{q}{p} \right) \equiv \left(\frac{p^*}{q} \right) \pmod{q}.$$

Finally,

$$\left(\frac{q}{p} \right) = \left(\frac{p^*}{q} \right) = \left(\frac{-1}{q} \right)^{\frac{p-1}{2}} \left(\frac{p}{q} \right) = \left((-1)^{\frac{q-1}{2}} \right)^{\frac{p-1}{2}} \left(\frac{p}{q} \right) = (-1)^{\frac{p-1}{2} \frac{q-1}{2}} \left(\frac{p}{q} \right).$$

§5 Miscellaneous Examples

5.1 Example (Putnam 1954)

Prove that there are no integers x and y for which

$$x^2 + 3xy - 2y^2 = 122.$$

PROOF

Multiplying by 4 and completing the square, we get $(2x + 3y)^2 - 17y^2 = 488$.

Since $488 \equiv 12 \pmod{17}$, it is enough to show that 12 is a quadratic nonresidue modulo 17.

By the *Quadratic Reciprocity Law*,

$$\left(\frac{12}{17}\right) = \left(\frac{3}{17}\right)\left(\frac{4}{17}\right) = \left(\frac{3}{17}\right) = \left(\frac{17}{3}\right) = \left(\frac{2}{3}\right) = (-1)^{\frac{3^2-1}{8}} = -1.$$

5.2 Example

Prove that there are infinitely many primes of the form $4n + 1$.

PROOF

Assume that $p_1, p_2, \dots, p_m$ are a finite set of such primes. Consider $Q = (2p_1 p_2 \dots p_m)^2 + 1$. If Q is prime then a prime of the form $4n + 1$ exists that is not among $p_1, p_2, \dots, p_m$ since $(2p_1 p_2 \dots p_m)^2 + 1$ leaves a remainder of 1 when divided by each of $p_1, p_2, \dots, p_m$. If Q is not prime, then an odd prime p divides Q .

Consequently, $\left(\frac{-1}{p}\right) = 1$ and thus p will be of the form $4n + 1$. Again p is not among the $p_1, p_2, \dots, p_m$ since $(2p_1 p_2 \dots p_m)^2 + 1$ leaves a remainder of 1 when divided by each of $p_1, p_2, \dots, p_m$. Thus it has been shown every finite set of primes of the form $4n + 1$ excludes some primes of that form. Therefore the set of such primes is infinite.

5.3 Example

Prove that there are infinitely many primes of the form $8n + 7$.

PROOF

Let $p_1, p_2, \dots, p_m$ be a finite set of such primes. Consider $Q = (4p_1 p_2 \dots p_m)^2 - 2$. An odd prime factor p of Q is either of the form $8n + 1$ or $8n + 7$, since 2 is a quadratic residue of Q . Not all odd prime factors can have the form $8n + 1$ because products of integers of the form $8n + 1$ are always of the form $8n + 1$ and Q has factor of the form $8n + 7$. Let p be a prime factor of the form $8n + 7$. Now p is not among the $p_1, p_2, \dots, p_m$ since for each i , $p_i \nmid (4p_1 p_2 \dots p_m)^2 - 2$. Thus it has been shown every finite set of primes of the form $8n + 7$ excludes some primes of that form. Therefore the set of such primes is infinite.

5.4 Example

Determine those odd primes p for which 3 is a quadratic residue and those for which it is a nonresidue.

SOLUTION

By the *Quadratic Reciprocity Law*,

$$\left(\frac{3}{p}\right) = (-1)^{\frac{(p-1)(3-1)}{4}} \left(\frac{p}{3}\right) = (-1)^{\frac{p-1}{2}} \left(\frac{p}{3}\right).$$

To determine $\left(\frac{p}{3}\right)$ we need to know the value $p \bmod 3$, and to determine $(-1)^{\frac{p-1}{2}}$ we need to know $p \bmod 4$.

Hence we need to consider $p \bmod 12$.

There are 4 cases:

(i). $p \equiv 1 \pmod{12}$:

Then $p \equiv 1 \pmod{3}$ so $\left(\frac{p}{3}\right) = \left(\frac{1}{3}\right) = 1$. Also $p \equiv 1 \pmod{4}$ so $(-1)^{\frac{p-1}{2}} = 1$.

Hence $\left(\frac{3}{p}\right) = 1$.

(ii). $p \equiv 5 \pmod{12}$:

Then $p \equiv 2 \pmod{3}$ so $\left(\frac{p}{3}\right) = \left(\frac{2}{3}\right) = (-1)^{\frac{3^2-1}{8}} = -1$. Also $p \equiv 1 \pmod{4}$ so $(-1)^{\frac{p-1}{2}} = 1$.

Hence $\left(\frac{3}{p}\right) = -1$.

(iii). $p \equiv 7 \pmod{12}$:

Then $p \equiv 1 \pmod{3}$ so $\left(\frac{p}{3}\right) = \left(\frac{1}{3}\right) = 1$. Also $p \equiv 3 \pmod{4}$ so $(-1)^{\frac{p-1}{2}} = -1$.

Hence $\left(\frac{3}{p}\right) = -1$.

(iv). $p \equiv 11 \pmod{12}$:

Then $p \equiv 2 \pmod{3}$ so $\left(\frac{p}{3}\right) = \left(\frac{2}{3}\right) = (-1)^{\frac{3^2-1}{8}} = -1$.

Also $p \equiv 3 \pmod{4}$ so $(-1)^{\frac{p-1}{2}} = -1$.

Hence $\left(\frac{3}{p}\right) = 1$.

Therefore, we summarize the above results and get

$$\left(\frac{3}{p}\right) = \begin{cases} 1, & \text{if } p \equiv \pm 1 \pmod{12}, \\ -1, & \text{if } p \equiv \pm 5 \pmod{12}. \end{cases}$$

5.5 Example

Determine those odd primes p for which 5 is a quadratic residue and those for which it is a nonresidue.

SOLUTION

By the *Quadratic Reciprocity Law*,

$$\left(\frac{5}{p}\right) = (-1)^{\frac{(p-1)(5-1)}{4}} \left(\frac{p}{5}\right) = \left(\frac{p}{5}\right).$$

Since the only quadratic residues of 5 are 1 and 4, it follows that

$$\left(\frac{5}{p}\right) = \begin{cases} 1, & \text{if } p \equiv \pm 1 \pmod{5}, \\ -1, & \text{if } p \equiv \pm 2 \pmod{5}. \end{cases}$$

5.6 Example

Show that there are no integers a, b for which $2b^2 + 3$ divides $a^2 - 2$.

PROOF

Observe that

$$2b^2 + 3 \equiv \begin{cases} 3 \pmod{8}, & \text{if } b \text{ is even,} \\ -3 \pmod{8}, & \text{if } b \text{ is odd.} \end{cases}$$

Hence $2b^2 + 3$ must have prime factors p with $p \equiv \pm 3 \pmod{8}$ otherwise all of its prime factors p will be of the form $p \equiv \pm 1 \pmod{8}$, which is impossible.

Now, for a given prime p with $p \equiv \pm 3 \pmod{8}$, we have $\left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}} = -1$.

Thus p does not divide $a^2 - 2$.

5.7 Example

Prove that if $m > 1$ and $n > 0$ have the same parity, then $3^n - 1$ is not divisible by $2^m - 1$

PROOF

If m is even, then $2^m = 4^{\frac{m}{2}} \equiv 1 \pmod{3}$. Hence $2^m - 1$ does not divide $3^n - 1$ for $n > 0$ regardless of the parity of n .

Suppose m and n are both odd with $m > 1$, and $2^m - 1$ divides $3^n - 1$. Since m is odd, it follows that $2^m - 1 \equiv 7 \pmod{12}$ (because $2^{m-2} - 2 \equiv 0 \pmod{3}$ and so $2^m - 8 \equiv 0 \pmod{12}$). Thus $2^m - 1$ has a prime factor p with $p \equiv \pm 5 \pmod{12}$ (otherwise all of its factors p are of the form $p \equiv \pm 1 \pmod{12}$ which is impossible). If p divides $3^n - 1$, then $3^{n+1} \equiv 3 \pmod{p}$. Since $n + 1$ is even, it follows that 3 is a quadratic residue modulo p . This is a contradiction because for $p \equiv \pm 5 \pmod{12}$, 3 is a quadratic nonresidue.

5.8 Example

Let p be a positive odd prime and let q denote the least positive quadratic nonresidue modulo p .

Then q is prime.

PROOF

The product of quadratic residues is a quadratic residue. So if q is not prime then one of its factors is a quadratic nonresidue modulo p which contradicts the fact that q is the least positive quadratic nonresidue modulo p .

5.9 Example

Let p be a positive odd prime and let q denote the least positive quadratic nonresidue modulo p .

Then $q < 1 + \sqrt{p}$.

PROOF

Let m be the least positive number for which $mq > p$, so that $(m - 1)q \leq p$. As $q \geq 2$ and p is prime, then $(m - 1)q < p$. Thus $mq - p < q$. As q is the least positive quadratic nonresidue modulo p , it follows

that $\left(\frac{mq - p}{p}\right) = 1$, and hence that $\left(\frac{m}{p}\right) = -1$. Consequently, $m \geq q$, so that

$$(q - 1)^2 < (q - 1)q \leq (m - 1)q < p.$$

5.10 Example

Let p be an odd prime. Evaluate $\sum_{r=1}^{p-2} \left(\frac{r(r+1)}{p}\right)$.

SOLUTION

For r with $\gcd(r, p) = 1$, there is an integer x such that $rx \equiv 1 \pmod{p}$.

Then

$$\left(\frac{r(r+1)}{p}\right) = \left(\frac{r(r+rx)}{p}\right) = \left(\frac{r^2(1+x)}{p}\right) = \left(\frac{1+x}{p}\right).$$

As r goes from 1 to $p - 2$, the least positive residue of $1 + x$ modulo p goes from 2 to $p - 1$. Thus the above sum is the same as

$$\sum_{r=2}^{p-1} \left(\frac{r}{p}\right) = -\left(\frac{1}{p}\right) = -1.$$

5.11 Example

Let a be a positive integer and let p be an odd prime such that $a < p$.

Denote by $N(a, p)$ the number of residues satisfying $\left(\frac{r}{p}\right) = \left(\frac{r+a}{p}\right) = +1$.

Find $N(a, p)$.

SOLUTION

First we evaluate the sum

$$S = \frac{1}{4} \sum_{r=0}^{p-1} \left(1 + \left(\frac{r}{p}\right)\right) \left(1 + \left(\frac{r+a}{p}\right)\right).$$

Multiplying out the summand, we get

$$S = \frac{1}{4} \left(p + \sum_{r=1}^{p-1} \left(\frac{r(r+a)}{p}\right) \right) = \frac{1}{4} \left(p + \sum_{r=1}^{p-1} \left(\frac{1+xa}{p}\right) \right) = \frac{p-1}{4}$$

where x is such that $rx \equiv 1 \pmod{p}$ and the last step uses the fact that $1 + xa$ goes through every nonzero value except 1 as r goes from 1 to $p - 1$. Consider the contribution of the r th term to S . If both r and $r + a$ are quadratic residues modulo p then the contribution is 1, and if either r or $r + a$ is a quadratic nonresidue the contribution is 0. This is almost $N(a, p)$.

Now observe that there are two terms we need to exclude, namely $r = 0$ and $r = p - a$. It follows that

$$\begin{aligned} N(a, p) &= \frac{1}{4} \left(p - 1 - \left(1 + \left(\frac{a}{p} \right) \right) - \left(1 + \left(\frac{-a}{p} \right) \right) \right) \\ &= \frac{1}{4} \left(p - 3 - \left(\frac{a}{p} \right) - \left(\frac{-a}{p} \right) \right). \end{aligned}$$

5.12 Example (IMO 1996, no.4)

The positive integers a and b are such that the numbers $15a + 16b$ and $16a - 15b$ are both squares of positive integers. What is the least possible value that can be taken on by the smaller of these two squares?

SOLUTION 1 Official Solution

Let $15a + 16b = r^2$ and $16a - 15b = s^2$, where r, s are positive integers.

Then

$$r^4 + s^4 = (15^2 + 16^2)(a^2 + b^2) = 481(a^2 + b^2).$$

Note that $481 = 13 \times 37$.

We now use the fact that -1 is not a fourth power either modulo 13 or modulo 37.

To see why this holds, note that the congruence $-1 \equiv x^4 \pmod{13}$, for some positive integer x , leads, via *Fermat's Little Theorem*, to

$$-1 = (-1)^3 \equiv x^{12} \equiv 1 \pmod{13},$$

which is false; likewise, the congruence $-1 \equiv x^4 \pmod{37}$, for some positive integer x , leads, via *Fermat's Little Theorem*, to

$$-1 = (-1)^9 \equiv x^{36} \equiv 1 \pmod{37},$$

which too is false.

Now, since

$$r^4 + s^4 \equiv 0 \pmod{13},$$

either

$$r \equiv s \equiv 0 \pmod{13}$$

or

$$r \not\equiv 0, s \not\equiv 0 \pmod{13}.$$

The latter possibility cannot occur because -1 is not a fourth power modulo 13.

Hence $r \equiv s \equiv 0 \pmod{13}$. Similarly, $r \equiv s \equiv 0 \pmod{37}$.

Therefore r and s are multiples of 481, and so $r \geq 481, s \geq 481$.

It is easy to check that $r = s = 481$ is realizable:

$$\text{take } a = 481 \times 31, b = 481.$$

Thus the required answer is 481^2 .

SOLUTION 2 Solution using *Quadratic Reciprocity Law*

Let $15a + 16b = r^2$ and $16a - 15b = s^2$, where r, s are positive integers. Then

$$15r^2 + 16s^2 = (15^2 + 16^2)a = 481a.$$

It follows that

$$-1 \not\equiv \left(\frac{16s^2}{13}\right) = \left(\frac{-15r^2}{13}\right) = \left(\frac{-15}{13}\right)\left(\frac{r^2}{13}\right)$$

and

$$-1 \not\equiv \left(\frac{16s^2}{37}\right) = \left(\frac{-15r^2}{37}\right) = \left(\frac{-15}{37}\right)\left(\frac{r^2}{37}\right).$$

By the *Quadratic Reciprocity Law*,

$$\begin{aligned} \left(\frac{-15}{13}\right) &= \left(\frac{-1}{13}\right)\left(\frac{3}{13}\right)\left(\frac{5}{13}\right) \\ &= 1 \cdot (-1)^{\frac{(3-1)(13-1)}{4}} \left(\frac{13}{3}\right) \cdot (-1)^{\frac{(5-1)(13-1)}{4}} \left(\frac{13}{5}\right) \\ &= \left(\frac{1}{3}\right)\left(\frac{3}{5}\right) \\ &= -1, \end{aligned}$$

and

$$\begin{aligned} \left(\frac{-15}{37}\right) &= \left(\frac{-1}{37}\right)\left(\frac{3}{37}\right)\left(\frac{5}{37}\right) \\ &= 1 \cdot (-1)^{\frac{(3-1)(37-1)}{4}} \left(\frac{37}{3}\right) \cdot (-1)^{\frac{(5-1)(37-1)}{4}} \left(\frac{37}{5}\right) \\ &= \left(\frac{1}{3}\right)\left(\frac{2}{5}\right) \\ &= -1. \end{aligned}$$

Hence

$$\left(\frac{r^2}{13}\right) \neq 1 \quad \text{and} \quad \left(\frac{r^2}{37}\right) \neq 1$$

so that $\left(\frac{r^2}{13}\right) = \left(\frac{r^2}{37}\right) = 0$. Thus $13 \mid r$ and $37 \mid r$. Since $13 \times 37 \mid 15r^2 + 16s^2$, it follows that $13 \mid s$ and

$37 \mid s$. Hence $481 \mid r$ and $481 \mid s$. Take $a = 481 \times 31$, $b = 481$. Then

$$15a + 16b = 481(15 \times 31 + 16) = 481^2,$$

$$16a - 15b = 481(16 \times 31 - 15) = 481^2.$$

Thus the required answer is 481^2 .

§6 Appendix

Lagrange's Theorem

Given a prime p , let

$$f(x) = a_0 + a_1 x + \dots + a_n x^n$$

be a polynomial of degree n with integer coefficients such that $a_n \not\equiv 0 \pmod{p}$.

Then the polynomial congruence $f(x) \equiv 0 \pmod{p}$ has at most n solutions.

PROOF

Use induction on n , the degree of f .

When $n = 1$, the congruence is linear:

$$a_0 + a_1 x \equiv 0 \pmod{p}.$$

Since $a_1 \not\equiv 0 \pmod{p}$, then $\gcd(a_1, p) = 1$ and there is exactly one solution.

Assume, then, that the theorem is true for polynomials of degree $n - 1$.

Assume also that the congruence $f(x) \equiv 0 \pmod{p}$ has $n + 1$ incongruent solutions modulo p , say,

$$r_0, r_1, \dots, r_n,$$

where $f(r_k) \equiv 0 \pmod{p}$ for each $k = 0, 1, \dots, n$.

Now, observe that

$$f(x) - f(r_0) = \sum_{k=1}^n a_k (x^k - r_0^k) = (x - r_0)g(x)$$

where $g(x)$ is a polynomial of degree $n - 1$ with integer coefficients and with leading coefficient a_n .

Thus

$$f(r_k) - f(r_0) = (r_k - r_0)g(r_k) \equiv 0 \pmod{p}$$

since $f(r_k) \equiv f(r_0) \equiv 0 \pmod{p}$.

But $r_k - r_0 \not\equiv 0 \pmod{p}$ if $k \neq 0$ so $g(r_k) \equiv 0 \pmod{p}$ for each $k \neq 0$.

This means that $g(x) \equiv 0 \pmod{p}$ has n incongruent solutions modulo p , contradicting the assumption.